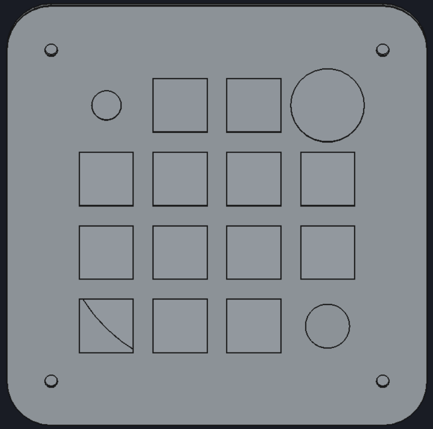
VibeKey Rev A

Two-device bill of materials

16 dynamically mapped inputs

Wireless + USB-C • 1,000 mAh • per-key RGB

Prototype procurement decision – 28 July 2026



What we are building

The BOM follows the current square enclosure and 16-input architecture

Controls

13 tactile MX keys
1 analog joystick + push
1 rotary encoder + push
1 capacitive touch region

Industrial design

114 × 114 mm square shell
6 mm corner radius
4° key-plane slope
86 mm circular bottom puck

User experience

Every input remappable
BLE and wired USB
13 individually addressed RGB keys
Clear caps + one ceramic accent

The supplied visual sketch is an aesthetic reference; the electrical count remains the previously specified 4×4 set of sixteen logical inputs.

Electronics selection

Exact primary components for each of the two devices

Compute + radio

Raytac MDBT50Q-1MV2
nRF52840 certified module

Power path

TI BQ24074 charger
TPS63031 3.3 V buck-boost

RGB power

TPS61023 5 V boost
TPS22918 switch + AHCT buffer

Sensors

Alps RKJX21224001 joystick
AT42QT1010 touch
MAX17048 fuel gauge

Connectivity

GCT USB4105 USB-C
TI USB ESD + Nexperia VBUS TVS

Battery

TinyCircuits ASR00012
Protected 1,000 mAh; 1 A max

Exactly two devices

No third assembled unit and no discretionary spare quantities

26 each

Kailh Polia switches
Kailh hot-swap sockets
1N4148W matrix diodes
12 mA SK6812MINI-E LEDs

2 each

Radio modules • batteries
USB-C • joystick • encoder
Every power and sensor IC
JST-SH battery connectors

Pack minimums

2 × Adafruit 5068 12-cap packs
1 × Cerakey four-cap pack
1 silicone sheet + adhesive
5 bare PCBs may be fab minimum

57 frozen BOM lines include per-board reference designators, fabricated parts, and reusable programming tools.

JLC attrition quantities are manufacturing consumables—not parts for a third device.

The machine-readable CSV is the purchasing authority for quantities.

Who assembles what

Designed around no at-home reflow

JLCPCB / contract manufacturer

All SMT passives and ICs
Raytac radio module
BQ24074 QFN charger
Power converters and protection
13 reverse-mounted RGB LEDs
13 hot-swap sockets + diodes
USB-C hybrid assembly preferred

Hand assembly

Install encoder and joystick
Insert MX switches and keycaps
Connect protected battery
Install heat-set inserts and screws
Fit printed enclosure parts
Cut/apply circular silicone pad
Program and run acceptance test

Power budget and non-negotiable limits

RGB budget

13 × 12 mA channels
≈169 mA total at 5 V incl. static
≈313 mA battery draw at 3.0 V
Firmware cap: 156 mA channels

Battery + charging

1,000 mAh protected cell
1 A maximum discharge
Target total draw <0.45 A
Charge limited to ≈0.4 A

Thermal constraint

BQ24074 may dissipate ≈0.8 W
Thermal copper/vias required
Closed-case charge test mandatory
No USB-PD claim

Do not substitute generic 60 mA-per-pixel LEDs.

The RGB rail defaults off and is enabled only after controlled startup.

Two-device prototype budget

Planning allowance; final price comes from live JLC and distributor carts

PCBs + assembly

$110–190
Five bare boards may be the fabrication minimum; populate only two. Includes double-sided setup and ordinary SMT labor.

Controls + mechanics

$90–170
Switches, caps, joysticks, encoders, batteries, hardware, printed parts, and foot material.

Shipping + contingency

$50–100
Multiple vendors, taxes, rare-part loading, and first-article rework allowance.

$250–460 total • $125–230 per working prototype

Before placing the complete order

The BOM is selected; product validation still depends on the PCB and physical prototype

- Complete schematic and four-layer routing; pass ERC/DRC and independent review.
- Replace every placeholder with its exact manufacturer footprint and STEP model.
- Redesign battery pocket, joystick opening/tail path, USB opening, and insert bosses.
- Verify radio antenna keep-out against copper, battery, screws, and circular base.
- Validate every JLC MPN, side, rotation, and package in the live order preview.
- Build two units; test power, charging temperature, USB, BLE range, RGB, touch, all controls, and enclosure fit.

Recommendation

Buy now

Two batteries for fit checks
Two joysticks and encoders
Two keycap packs + ceramic pack
Switches and mechanical samples
Programming cable / nRF52840-DK if needed

Hold until layout validation

JLC assembled PCB order
Final printed enclosure batch
Custom-cut production foot pads
Any volume purchase beyond the two devices

Procurement source: hardware/bom.csv

Engineering rationale and gates: hardware/BOM.md